

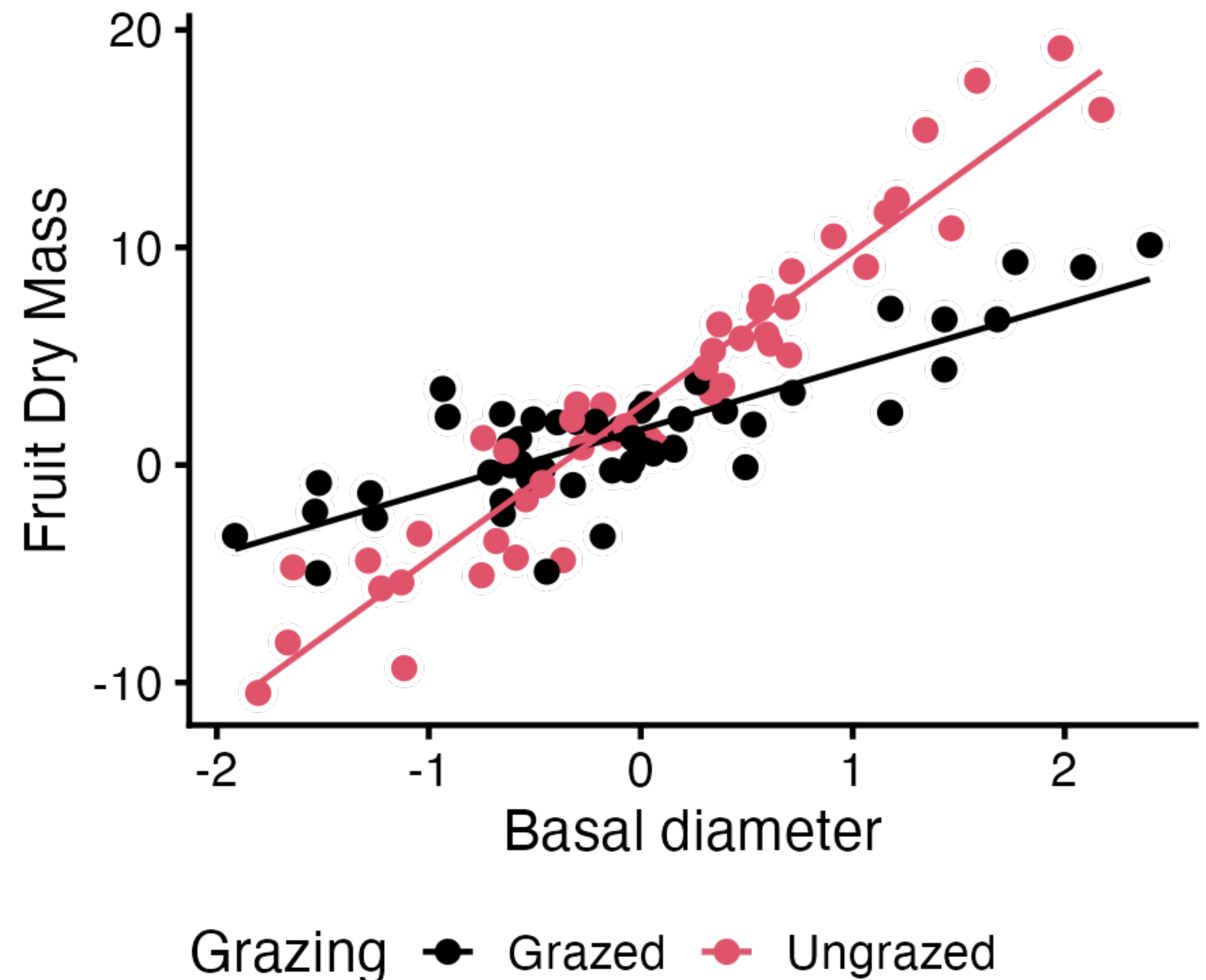
Interactions

Allowing coefficients to change according to other variables

- What if the relation between a predictor and an outcome changes depending on another predictor?
- We can account for this by adding product terms in the model
- With two predictors, x and z :

$$\mu = \alpha + \beta_1 x + \beta_2 z + \beta_3 xz$$

Interaction term!



Example with interactions

- **Question:** what are the best soil moisture and light availability to grow tulips?
- Greenhouse experiment: beds of tulips kept in nine combinations of soil moisture and shading. Three replicates for each combination (total of 27 beds).
- **Response variable:** mean plant height in each bed
- **Predictor variables:**
 - Water: 3 levels (low, med, high)
 - Shade: 3 levels (low, med, high)

